Special Notice

Information Innovation Office (I2O)

Exponentiating Mathematics (expMath) Proposers Day

DARPA-SN-25-44

March 25, 2025

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Opportunity	Exponentiating Mathematics (expMath)
Event	Hybrid Proposers Day
Event Date and Time	4/23/25 at 10:00 AM Eastern Time (ET)
Event Physical Location	DARPA Conference Center 675 N Randolph St Arlington, VA 22203
Fee to Attend	There is **NO** fee to attend
Who can Attend	This event is open to potential proposers only; not open to the public.
Registration Website	https://creative.spa.com/darpa/i2o/expmath/pd/
Registration Deadline	4/16/25 at 5:00 PM (ET)
Non-U.S. Registration Forms Deadline	4/8/25 at 5:00 PM (ET)
Administrative POC	expMath@darpa.mil
DARPA Program Manager	Dr. Patrick Shafto

PROPOSERS DAY

DARPA I2O is sponsoring an unclassified hybrid Proposers Day in support of the anticipated Exponentiating Mathematics (expMath) Broad Agency Announcement (BAA). The purpose of this event is to provide information on the expMath technical goals and challenges, address questions from potential proposers, and provide an opportunity for potential proposers to consider how their research may align with the expMath program objectives. Attendance is voluntary and is not required to propose to the expMath BAA.

Program Background and Objectives

Mathematics is the source of significant technological advances; however, progress in math is slow. Recent advances in artificial intelligence (AI) suggest the possibility of increasing the rate of progress in mathematics. Still, a wide gap exists between state-of-the-art AI capabilities and pure mathematics research.

Advances in mathematics are slow for two reasons. First, decomposing problems into useful lemmas is a laborious and manual process. To advance the field of mathematics, mathematicians use their knowledge and experience to explore candidate lemmas, which, when composed together, prove theorems. Ideally, these lemmas are generalizable beyond the specifics of the current problem so they can be easily understood and ported to new contexts. Second, proving candidate lemmas is slow, effortful, and iterative. Putative proofs may have gaps, such as the one in Wiles' original proof of Fermat's last theorem, which necessitated more than a year of additional work to fix. In theory, formalization in programming languages, such as Lean, could help automate proofs, but translation from math to code and back remains exceedingly difficult.

The significant recent advances in AI fall short of the automated decomposition or auto(in)formalization challenges. Decomposition in formal settings is currently a manual process, as seen in the Prime number theorem and beyond and the Polynomial Freiman-Ruzsa conjecture, with existing tools, such as Blueprint for Lean, only facilitating the structuring of math and code. Auto(in)formalization is an active area of research in the AI literature, but current approaches show poor performance and have not yet advanced to even graduate-level textbook problems. Formal languages with automated theorem-proving tools, such as Lean and Isabelle, have traction in the community for problems where the investment in manual formalization is worth it.

The goal of expMath is to radically accelerate the rate of progress in pure mathematics by developing an AI co-author capable of proposing and proving useful abstractions. expMath will be comprised of teams focused on developing AI capable of auto decomposition and auto(in)formalization and teams focused on evaluation with respect to professional-level mathematics. We will robustly engage with the math and AI communities toward fundamentally reshaping the practice of mathematics by mathematicians. The expMath program will be comprised of two technical areas (TAs). TA1 will focus on automation of decomposition and formalization/informalization. TA2 will focus on evaluation of progress relative to professional mathematics, as represented by research papers and graduate level textbooks.

TA1 potential proposers are expected to have strong expertise in ***both*** mathematics and AI. TA2 potential proposers are expected to have strong expertise in ***both*** designing evaluations of AI technologies and experience in mathematics.

For more information, see the expMath BAA at https://sam.gov/.

Registration Information

The expMath hybrid Proposers Day will be held on 4/23/25, from 10:00 AM to 4:00 PM (ET) physically at the DARPA Conference Center, located at 675 N Randolph St., Arlington, Virginia, 22203 and virtually via Zoom for Government. No more than three representatives per organization unit, and five people per overall organization, may attend this Proposers Day. Registrants are strongly encouraged to coordinate attendance internally within their organizations prior to registration.

In-person Registration

In-person registration must be completed no later than 4/16/25 at 5:00 PM ET. Registration is on a first come first serve basis with a limit of 176 participants. Any United States (U.S.) citizen seeking to attend the Proposers Day in-person will be required to present a valid, Government-issued photo identification at the registration table prior to admittance into the event.

Late in-person registrations may be admitted after all pre-registered attendees are admitted, and only if room capacity allows. Please note, in-person check-in begins at 8:00 AM ET; late in-person registration requires significant coordination and may not be complete in time to gain physical entrance into the Proposers Day event.

Virtual Registration

Virtual registration must be completed no later than 4/16/25 at 5:00 PM ET. All virtual attendees are required to log in using their first and last name. Participants calling in must call from the number provided during registration.

Non-U.S. Registration

Any **non-U.S.** citizen must complete and submit one of the following:

- DARPA Form 60; or
- Foreign government personnel must submit an Official Visit Request completed by their respective Embassy based in Washington, DC.

All **non-U.S.** forms must be submitted no later than 4/8/25 at 5:00 PM ET. Form 60 submission instructions are provided on the registration website and in the registration confirmation email. Foreign government personnel should contact their respective Embassy staff for assistance in submitting the Official Visit Request. All non-U.S. personnel must present a valid passport or Permanent Resident Card.

Event Format

The expMath Proposers Day will include overview presentations by the DARPA expMath Program Manager, Dr. Patrick Shafto, and DARPA support offices. The event will conclude with a Question & Answer (Q&A) session.

Following the Proposers Day, DARPA anticipates posting additional information regarding the expMath (e.g., DARPA materials presented at the Proposers Day and a Q&A document) to:

http://www.darpa.mil/work-with-us/opportunities.

ATTENDEE TEAMING

To assist those wanting to establish collaborative teaming efforts and business relationships, potential proposers may indicate on the registration website that they authorize their contact information to be shared among meeting participants. For persons or organizations new to expMath, this may be of use in connecting with potential partners. The expMath team anticipates potential proposers will have the opportunity to conduct informal teaming discussions in-person and virtually during and after the event. The expMath team may provide informal communication channels to assist in teaming discussions during and after the event. Potential proposers are strongly encouraged not to rely solely on this event or potential government communication channels to form teaming relationships.

NOTE: This Special Notice does not constitute a formal solicitation for proposals. This notice is issued solely for information and expMath planning purposes and is not a Request for Information (RFI). Since this is not an RFI, DARPA will not accept submissions against this notice. DARPA will not provide reimbursement for costs incurred to participate in this Proposers Day. Interested parties to this notice are cautioned that nothing herein obligates DARPA to issue a solicitation.